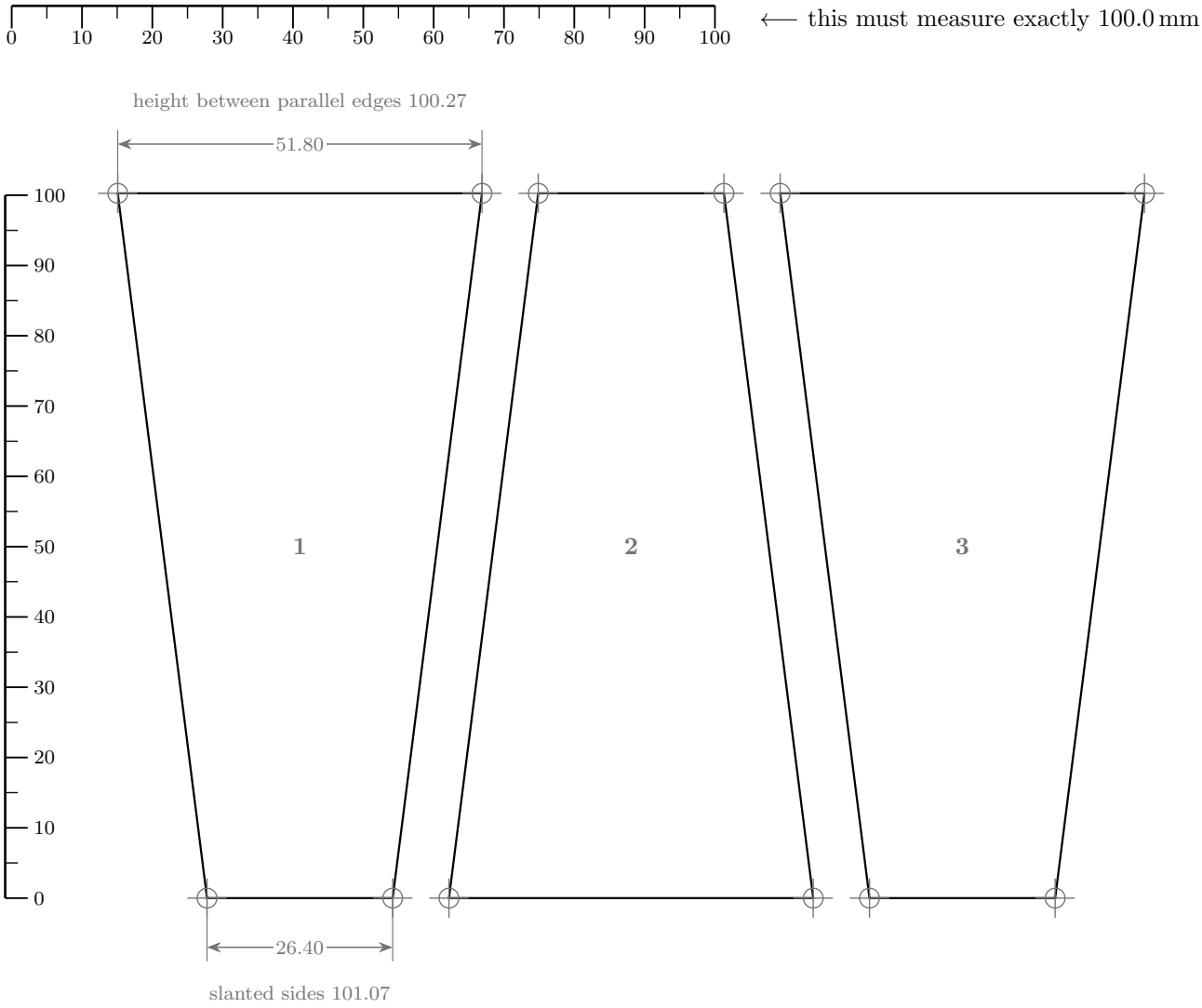
Kaleidoscope mirror cutting template

100 mm prototype · three identical mirrors · drawn 1:1

Print at 100%. In the print dialog choose *Actual size* or *100%* — not *Fit to page*, not *Shrink oversized pages*. Then check *both* rulers below with a real ruler before you cut anything. If they are off, so is every dimension on this page.



How to use it. Leave the protective film on the mirror. Tape this sheet down over it, prick through the centre of all twelve circled marks with a pin or a scribe, lift the paper off, then join the pricks with your straightedge and score along those lines. Marking through the paper is more accurate than trying to cut through it — a knife shreds paper and then wanders.

These blanks are 1.00 mm wider than the optical face, which is 25.40 and 50.80 mm. The surplus is deliberate. At assembly each mirror is pushed toward one corner until its leading edge stops against the neighbour's mirror, and the surplus buries itself in the dead space behind — so a blank that comes out a little wide or a little narrow still seats with no gap at the seam.

Anything from 25.5 to 27.4 mm at the narrow end will work, measured across the parallel edge. Err generous and sand back rather than cutting tight: too narrow eventually opens a gap at the seam, which prints as a dark line across every facet, while too wide is harmless until the surplus runs out of dead space.

Cut with the coated side facing up. These are rear-surface mirrors — clear plastic in front, reflective coating on the back. Blade down through the coating gives a clean coating edge; the tear-out from the last pass then lands on the clear face, where it does not matter. Slip a sheet of paper under the clear face so the mat cannot mark it.

Only the slanted edges matter. The 101.07 mm sides butt against the neighbouring mirrors — they must be straight, and the reflective coating must run right to the edge. The 26.40 and 51.80 mm ends are hidden under the end caps and can be rough.

There is no left or right. Each trapezoid is symmetric, so all three pieces are interchangeable and either slanted edge can end up as the overlap. What must be consistent is the *assembly*: all three mirrors get pushed the same way round. Mirror

2 is drawn flipped only to nest the three efficiently on the tile.